Electoral Lens — Indian General Election 2024 Analytics

Technology Stack:
Python • Pandas • MySQL • SQL • Power BI • DAX

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EXECUTIVE SUMMARY

The **Indian General Election 2024 Analytics** project is an end-to-end data analytics and business intelligence solution developed to analyze the electoral outcomes of the 2024 Lok Sabha Election. The project focuses on transforming election data into meaningful analytical insights covering party performance, alliance-level results, geographical distribution, candidate victories, gender representation, age distribution and independent candidates.

The project uses three structured datasets — **candidates_with_phase.csv, results_2024.csv and results_2024_winners.csv** — containing candidate information, constituency-level voting results and winning-candidate information. The datasets cover the complete election across **543 constituencies, 36 States/Union Territories, 746 parties and seven election phases**. The cleaned analytical data was validated to ensure consistency across candidates, constituencies, parties and election outcomes.

Python and Pandas were used as the initial data-processing layer. The datasets were inspected, standardized and transformed into analysis-ready CSV files. Data-quality checks included structural inspection, missing-value analysis, duplicate checks, data-type handling, text standardization, date parsing and validation of important election entities. Special care was taken with election-specific values such as uncontested results and missing vote fields rather than treating every missing value as zero.

The processed data was organized into a relational MySQL database and analyzed using SQL. The database architecture separated important election entities such as states, regions, constituencies, phases, parties, candidates, results and winners. SQL joins were then used to create analytical datasets combining constituency, candidate, party, alliance and winner information.

The final Power BI solution consists of **10 dashboards**, covering Election Overview, Geographical Analysis, Party Performance, Statewise Results, Candidate Analysis, Independent Candidates and Election Key Insights. Interactive slicers for dimensions such as **State, Region, Alliance, Election Phase, Constituency and Party** allow users to explore the election dynamically.

The final solution provides a consolidated analytical view of the 2024 Lok Sabha Election and demonstrates how raw election data can be converted into a structured decision-support system. Instead of presenting only static election statistics, the project connects data preparation, relational database design, SQL analysis and interactive business intelligence into one complete analytical workflow.

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Introduction

The **Indian General Election 2024 Analytics** project is a comprehensive data analytics and business intelligence solution focused on understanding the outcome of the 2024 Lok Sabha Election.

The project belongs to the **Political and Electoral Data Analytics** domain and analyzes election results from multiple perspectives rather than focusing only on the final seat count.

The analysis covers:

- Overall Lok Sabha election results
- Alliance-wise seat distribution
- Party-wise seats and vote share
- Geographical distribution of results
- Statewise party performance
- Winning margins
- Closest electoral contests
- Gender representation
- Age distribution of elected MPs
- Youngest and oldest elected MPs
- Independent candidates and winners
- Election-level key insights

The project was developed to demonstrate how a large real-world dataset can be taken through a complete analytics pipeline — from data preparation and database organization to SQL analysis and interactive Power BI reporting.

The final solution therefore combines **data engineering, SQL analytics and business intelligence** rather than treating the dashboard as an isolated component.

Business Problem and Objectives

Business Problem

Election datasets contain a large amount of information distributed across candidates, parties, constituencies, states, alliances and election phases.

Looking only at the final seat count does not provide sufficient understanding of:

- Which parties performed strongly
- How alliances performed
- How electoral results varied geographically
- How large or narrow victory margins were
- How elected representatives were distributed by gender and age
- How independent candidates performed
- Which candidates achieved exceptional victories

The project addresses this problem by creating a structured analytical system that converts election-level data into interactive and interpretable insights.

Project Objectives

The main objectives were to:

- Analyze the overall 2024 Lok Sabha election outcome.
- Compare NDA, I.N.D.I.A. and Others.
- Analyze party-wise seats won.
- Analyze national party vote share.
- Study geographical distribution of electoral outcomes.
- Analyze statewise party performance.
- Identify candidates with the largest victory margins.
- Identify closely contested constituencies.
- Analyze gender representation among elected MPs.
- Analyze the age distribution of elected MPs.
- Identify the youngest and oldest elected MPs.
- Analyze independent candidates and independent winners.
- Build an interactive Power BI reporting solution.

- Provide a consolidated set of election-level insights.

Key Business Questions

The project answers questions such as:

- Which party won the highest number of Lok Sabha seats?
- Which alliance secured the largest number of seats?
- Which parties received the highest vote share?
- How are election results geographically distributed?
- Which parties dominated different states?
- Which candidates recorded the largest victory margins?
- Which constituencies experienced the closest victories?
- How many elected MPs were male and female?
- What is the age distribution of elected MPs?
- Who were the youngest and oldest elected MPs?
- How many independent candidates participated?
- How many independent candidates won?
- What are the most important overall findings from the 2024 election?

Technology Stack, Dataset and Project Architecture

Technology Stack

Python :	Data loading, inspection, cleaning, transformation and validation
Pandas :	DataFrame-based preprocessing and data manipulation
MySQL:	Relational database storage and structured analytical querying
SQL:	Joins, aggregations and creation of analytical datasets
Power BI :	Interactive dashboards and visualization
DAX :	Analytical measures and dashboard calculations

Dataset Overview

Dataset Source: Kaggle
Dataset: *Lok Sabha 2024 Results & Candidates*
The project uses the Lok Sabha 2024 Results & Candidates dataset available on Kaggle. The dataset contains three primary files — candidates_with_phase.csv, results_2024.csv, and results_2024_winners.csv — covering candidate information, constituency-level results, voting data and winning-candidate information.

The final project uses three primary CSV datasets:

candidates_with_phase.csv	Candidate-level information including election phase and candidate attributes
results_2024.csv	Constituency-level candidate voting results
results_2024_winners.csv	Winner, runner-up and victory-margin information

The candidate dataset contained **8,441 records and 14 columns**, the results dataset contained **8,902 records and 10 columns**, and the winners dataset contained **543 records and 9 columns** before the final analytical transformations.

The validated analytical dataset covered:

- **36 States/Union Territories**
- **543 constituencies**
- **746 parties**
- **8,101 candidates**
- **7 election phases**

Project Architecture

The project follows an end-to-end analytical pipeline:

Raw Election CSV Files
↓
Python / Pandas Data Inspection
↓
Data Cleaning & Transformation
↓
Validation & Clean CSV Files
↓
MySQL Relational Database
↓
SQL Joins & Analytical Queries
↓
Power BI Data Model
↓
DAX Measures & Calculations
↓
Interactive Dashboards
↓
Election Insights

The architecture separates data preparation, database management, analysis and visualization, making the project more suitable for a real-world analytics workflow.

Data Preparation, Analysis and Methodology

The data preparation stage was performed before building the final dashboards.

Data Loading and Inspection

The datasets were loaded into Python using Pandas and inspected to understand their structure.

The preprocessing workflow included:

- Reading the source CSV files
- Checking dataset shape
- Inspecting column names
- Reviewing data types
- Checking missing values
- Checking duplicates
- Examining unique values
- Identifying inconsistencies in categorical fields
- Reviewing numerical vote fields
- Inspecting date-related information

Data Standardization

The datasets were standardized to make them suitable for database loading and analysis.

This included:

- Standardizing column names
- Converting fields to appropriate data types
- Correcting inconsistent naming
- Cleaning candidate and party-related text
- Handling mixed date formats
- Converting numerical election fields appropriately

The project also preserved meaningful NULL values where a missing value represented the actual election situation rather than an ordinary data error.

For example, uncontested election situations were not blindly treated as ordinary zero-vote observations.

Data Validation

The cleaned datasets were validated against important election entities.

The final validation covered:

- State/UT count
- Constituency count
- Party count
- Candidate count
- Election phases
- Duplicate records
- Candidate and result consistency

MySQL Database

The cleaned data was then structured in MySQL.

The database separates election entities so that information such as party, constituency, candidate and result is not unnecessarily duplicated.

The analytical structure included entities for:

- States
- Regions
- Constituencies
- Parties
- Candidates
- Election phases
- Results
- Winners

The finalized SQL workflow also combined information from state, region, constituency, phase, winner summary, party, result and application-level entities to create a master analytical dataset.

SQL Analysis

SQL was used to combine and analyze the relational data.

The analysis involved:

- Joining related election entities
- Aggregating seats
- Calculating party-level performance

- Combining winner and runner-up information
- Comparing vote totals
- Analyzing winning margins
- Producing datasets required for Power BI

This ensured that the dashboard was supported by structured analytical data rather than relying entirely on raw CSV files.

Power BI Development

The processed analytical data was brought into Power BI.

The dashboard layer used:

- Interactive slicers
- KPI cards
- Bar charts
- Donut charts
- Maps
- Tables/button-style selectors
- Age-distribution charts
- Candidate comparison charts
- Text-based insight panels
- DAX measures

The final dashboard design consistently uses a **dark brown background with gold/yellow borders, headings and visual containers**, creating a unified visual identity across the project.

Election Overview and Geographical Analysis

Purpose

This component establishes the overall electoral picture before moving into detailed party, state and candidate analysis.

It contains:

- **Dashboard 1 – Election Overview**
- **Dashboard 2 – Geographical Analysis**

Dashboard 1 – Election Overview

The Election Overview dashboard provides a high-level summary of the 2024 Lok Sabha Election.

Key KPIs

The dashboard highlights:

- **543 Lok Sabha Seats**
- **293 NDA Seats**
- **234 I.N.D.I.A. Seats**
- **16 Others Seats**

Alliance Wise Seats

The donut chart compares:

- NDA — **293 seats**
- I.N.D.I.A. — **234 seats**
- Others — **16 seats**

This provides an immediate understanding of alliance-level electoral performance.

Interactive Filters

The dashboard includes:

- Region
- State
- Alliance

These filters allow users to move from the national election picture toward specific geographical or alliance-level views.

Analytical Question Answered

How were the 543 Lok Sabha seats distributed among the major alliances?

Dashboard 2 – Geographical Analysis

The Geographical Analysis dashboard provides a spatial view of constituency-level electoral outcomes.

Constituency Wise Results

The map displays constituencies across India according to alliance classification.

The visual allows users to identify geographical patterns in NDA, I.N.D.I.A. and Others performance.

Filters

The dashboard provides:

- State
- Region
- Alliance
- Election Phase
- Constituency

KPI Cards

The dashboard also displays:

- Total Seats — **543**
- NDA Seats — **293**
- I.N.D.I.A. Seats — **234**
- Others Seats — **16**

Analytical Question Answered

How are electoral outcomes geographically distributed across India and how does this distribution change when users filter by state, region, alliance or election phase?

Party and State Performance Analysis

This component focuses on party-level and state-level electoral performance.

It contains:

- **Dashboard 3 – Party Performance Analysis 1**
- **Dashboard 4 – Party Performance Analysis 2**
- **Dashboard 5 – Statewise Result Analysis**

Dashboard 3 – Party Performance Analysis 1

Party Wise Seats Won

The horizontal bar chart ranks parties according to seats won.

Major results include:

- Bharatiya Janata Party — **240**
- Indian National Congress — **99**
- Samajwadi Party — **37**
- All India Trinamool Congress — **29**
- Dravida Munnetra Kazhagam — **22**
- Telugu Desam — **16**

The visual allows the user to immediately identify the strongest parties.

Winning Party KPIs

The dashboard shows:

- **42 Total Winning Parties**
- **15 NDA Winning Parties**
- **20 I.N.D.I.A. Winning Parties**
- **7 Others Winning Parties**

Filters

- State
- Region
- Alliance

Analytical Question Answered

Which parties won seats and how widely distributed was winning performance across alliances?

Dashboard 4 – Party Performance Analysis 2

Party Wise Vote Share

The dashboard ranks parties according to national vote percentage.

The leading parties include:

- Bharatiya Janata Party — **36.81%**
- Indian National Congress — **21.33%**
- Samajwadi Party — **4.61%**
- All India Trinamool Congress — **4.40%**

Total Votes Polled

The dashboard reports approximately:

64.54 Crore Total Votes Polled

This provides an indication of the scale of electoral participation.

Filters

- State
- Alliance

Analytical Question Answered

Which parties received the highest national vote share, and what was the overall scale of voter participation?

Dashboard 5 – Statewise Result Analysis

This dashboard moves from national party performance to state-level performance.

Seats Won Statewise by Party

The donut chart displays the distribution of seats among winning parties.

It highlights the dominance of major parties while also showing the contribution of smaller parties.

State / UT Selector

The dashboard provides state-level selection buttons covering the States/UTs represented in the dataset.

This allows the analysis to move from the overall election result to specific geographical areas.

Analytical Question Answered

How does party performance vary across different States and Union Territories?

Candidate and Representation Analysis

This component focuses on individual candidates and elected-member characteristics.

It contains:

- **Dashboard 6 – Candidate Analysis 1**
- **Dashboard 7 – Candidate Analysis 2**
- **Dashboard 8 – Candidate Analysis 3**

Dashboard 6 – Candidate Analysis 1

Big Victories Margin Wise

The dashboard identifies candidates who achieved exceptionally large victory margins.

Shankar Lalwani recorded the largest victory margin at approximately **11.75 lakh votes**.

Other major high-margin winners are also displayed for comparison.

Neck to Neck Fight Victories

The second visual identifies closely contested constituencies.

Ravindra Dattatray Waikar recorded a victory margin of only **48 votes**, making it one of the closest victories in the election.

Filters

- State
- Alliance

Analytical Questions Answered

Which candidates recorded the largest victory margins?

Which candidates won by exceptionally narrow margins?

Dashboard 7 – Candidate Analysis 2

Gender Details

The dashboard analyzes gender representation among elected MPs.

The distribution shown is:

- Male — **469 MPs (86.37%)**
- Female — **74 MPs (13.63%)**

The visual demonstrates the significant difference between male and female representation among elected MPs.

Alliance Filter

The dashboard allows users to examine gender representation through:

- I.N.D.I.A.
- NDA
- Others

Political Party Filter

The dashboard also provides a party-based selection mechanism to examine the representation of elected MPs across political parties.

Analytical Question Answered

How is gender representation distributed among elected MPs across alliances and political parties?

Dashboard 8 – Candidate Analysis 3

This dashboard focuses on the age profile of elected MPs.

Age Group Distribution

The elected MPs are distributed across five age groups:

Age Group	MPs
Below 35	23
35–44	72
45–54	156
55–64	172
65+	120

The distribution shows that the largest group of elected MPs falls within the **55–64 age category**.

Youngest MP Elected

Priya Saroj

- Age: **25**
- Party: Samajwadi Party
- State: Uttar Pradesh

- Constituency: Machhlishahr

Oldest MP Elected

T. R. Baalu

- Age: **82**
- Party: Dravida Munnetra Kazhagam
- State: Tamil Nadu
- Constituency: Sriperumbudur

Filters

- Alliance
- State
- Region

Analytical Questions Answered

What is the age distribution of elected MPs?

Who was the youngest elected MP?

Who was the oldest elected MP?

Independent Candidates and Election Key Insights

This component combines the project's dedicated independent-candidate analysis with the final election-level storytelling dashboard.

It contains:

- **Dashboard 9 – Independent Candidates Analysis**
- **Dashboard 10 – Election Key Insights**

Dashboard 9 – Independent Candidates Analysis

Statewise Distribution of Independent Candidates

The donut chart displays the distribution of successful independent candidates across states.

The dashboard highlights independent winners from states/UTs including:

- Punjab
- Bihar
- Dadra & Nagar Haveli and Daman & Diu
- Jammu & Kashmir
- Ladakh
- Maharashtra

Winning Margin of Successful Independent Candidates

The bar chart compares the winning margins of successful independent candidates.

Independent Candidate KPIs

The dashboard shows:

7 Independent Winners

and

3,920 Total Independent Candidates

This distinction is important because the dashboard measures both participation and successful independent representation.

Analytical Questions Answered

How many independent candidates participated in the election?

How many independent candidates won?

Where were successful independent candidates located?

What were the winning margins of successful independent candidates?

Dashboard 10 – Election Key Insights

The final dashboard acts as the **storytelling and conclusion layer** of the Power BI solution.

It consolidates the most important findings from the previous dashboards.

Key Business Insights

The completed analysis provides several important insights into the 2024 Lok Sabha Election.

Strong Single-Party Performance

The **Bharatiya Janata Party won 240 seats**, making it the largest individual party in the election. The Indian National Congress followed with 99 seats.

This demonstrates a strong concentration of seats among the two largest national parties.

Alliance-Level Competition

NDA secured **293 seats**, while I.N.D.I.A. secured **234 seats** and Others secured **16 seats**.

The alliance-level view provides a more meaningful picture of parliamentary strength than examining parties independently.

BJP's Vote Share Leadership

BJP recorded the highest national vote share at **36.81%**, followed by Congress at **21.33%**.

This demonstrates the party's significant national electoral footprint.

Large-Scale Electoral Participation

Approximately **64.54 crore votes** were polled.

The figure highlights the enormous scale of the Indian electoral system and the importance of scalable data analytics for understanding election results.

Very Large Victory Margins

Shankar Lalwani recorded the largest victory margin at approximately **11.75 lakh votes**.

This demonstrates that electoral competitiveness varies considerably across constituencies.

Extremely Close Electoral Contests

At the opposite end, **Ravindra Dattatray Waikar** won by only **48 votes**.

The contrast between the largest and smallest victory margins demonstrates why candidate-level analysis is important alongside party-level analysis.

Gender Representation Gap

The dashboard shows **469 male MPs and 74 female MPs**.

Female representation therefore remains substantially lower than male representation, highlighting the need to promote greater women's representation and 33% reservation.

Age Diversity Among Elected MPs

The elected MPs span multiple age categories, with the largest group falling within the **55–64** age range.

The analysis also identifies a significant age range between the youngest and oldest elected representatives.

Youngest and Oldest Elected MPs

Priya Saroj, at **25 years**, was identified as the youngest elected MP in the analysis.

T. R. Baalu, at **82 years**, was identified as the oldest.

This demonstrates the wide generational range represented within the elected Lok Sabha.

Independent Participation

The analysis identified **3,920 independent candidates**, of whom **7 became winners**.

This provides a useful perspective on the scale of independent participation compared with successful independent representation.

Conclusion

The **Indian General Election 2024 Analytics** project successfully developed a complete end-to-end analytical solution for examining the 2024 Lok Sabha Election.

The project began with structured election datasets and progressed through **Python-based data preparation, Pandas preprocessing, validation, relational database organization in MySQL, SQL-based analysis and Power BI dashboard development**.

The resulting Power BI solution contains **10 dashboards**, moving progressively from the overall election picture to geographical analysis, party performance, statewise results, candidate outcomes, representation, independent candidates and finally a consolidated election-insights dashboard.

The project demonstrates the ability to work with a complex real-world dataset containing multiple entities and relationships. Instead of relying on a single visualization, the analysis connects party, alliance, state, constituency and candidate-level information to provide a broader understanding of election outcomes.

The final solution also demonstrates important Data Analyst skills: **data cleaning, data validation, relational thinking, SQL analysis, analytical modeling, DAX, interactive visualization, KPI development and business storytelling**.

Most importantly, the project converts raw electoral records into an interactive decision-support system where users can explore the election from multiple perspectives and quickly identify meaningful patterns.

Final Project Statement

The Indian General Election 2024 Analytics project demonstrates how an end-to-end data analytics pipeline can transform complex electoral data into an interactive business intelligence solution that explains not only who won, but also how parties, alliances, states and individual candidates shaped the overall election outcome.